

Homework 1.1 - Industry Structured & Unstructured Data

Sunday, October 26, 2025

6:13 PM

Many modern engineering systems must handle both structured and unstructured data to function effectively.

- **Choose an industry or application area such as healthcare, automotive, or finance.**

Quick Commerce

- **Describe one example of structured data and one example of unstructured data generated or used in that domain.**

Structured data: Inventory details

Semi-structured data: Address of shipping stores/consumers

Unstructured data: Product images/Customer requests

- **Explain how each type of data is stored and processed technically (e.g., databases, data lakes, big data tools).**

- Inventory details can be stored in relational databases (Data warehouse) and processed using queries or ETL pipelines
- Unstructured data like images can be stored in cloud object storage systems (Data Lakes) and processed using computer vision models along with parallel image processing in *Hadoop* or *Spark*

- **Discuss the engineering challenges involved in integrating and analyzing these two types of data to produce useful insights or automation.**

- Structured and unstructured data in different formats require different modes of storage
- Unlike structured, unstructured data usually requires some advanced NLP or image recognition pipeline for further processing making integration difficult
- Traditional BI tools which generally work well with structured data fall short in extracting deep meaning/insights from unstructured data

- **Suggest technologies, algorithms, or architectures that can help overcome these challenges (e.g., SQL, NoSQL, Hadoop, etc.).**

- Store structured in SQL (*PostgreSQL*, *Snowflake*) and semi and unstructured data in NoSQL databases (*Mongo*, *Cassandra*)
- Use data lakes like *S3* or *Hadoop HDFS* to store both types of data side by side
- Parallel processing of unstructured data like images/text can be made easy using *MapReduce* or *Spark*
- A data pipeline can be set up using *Talend* to automate collection, cleaning and integration of all types of data